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SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1

Case Study

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA0405**
SLOT: D

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%
Assessment Tool Weightage: 25%

Duration: 90 minutes
Marks: 25

Code of Conduct:

I **Ch. Venkata Lakshmi Hymavathi (192425397)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ch. Hymavathi

Name of the student:
Ch. Venkata Lakshmi Hymavathi

Criteria	Conceptual Understanding (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity & Presentation *(1)*	Total (10)
Weightage	*25 %*	*25%*	*20%*	*20 %*	*10 %*	**100%**
Q1						
Q2						
Total						

Signature of the Faculty: ***Total Marks :***

QUESTIONS: **Total Marks: 20**

Case Study Question 1: CNN for Automated Medical Image Classification (10 Marks)
A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

Question:

1. Design and explain a suitable **CNN architecture** for this medical image classification system.

   Answer:
   A suitable CNN can be designed as follows:
   Input Image → Convolution → ReLU → Pooling → Convolution → ReLU → Pooling → Convolution → ReLU → Flatten → Fully Connected → Dropout → Output
   For example:

- **Input:** Chest X-ray resized to 224 × 224 × 3.
- **Conv Layer 1:** 32 filters of 3×3 → ReLU → Max Pooling.
- **Conv Layer 2:** 64 filters of 3×3 → ReLU → Max Pooling.
- **Conv Layer 3:** 128 filters of 3×3 → ReLU → Max Pooling.
- **Flatten:** Converts feature maps into a vector.
- **Fully Connected Layer:** Learns the relationship between extracted features and classes.
- **Dropout:** Reduces overfitting.
- **Output Layer:** Two neurons/classes — Normal and Disease-Affected, using Softmax or Sigmoid.
  The network learns simple features in early layers and more complex medical patterns in deeper layers.

2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.

   Answer:
   CNNs are suitable because chest X-rays contain important spatial and visual patterns. CNNs automatically learn these features without manually designing them.

   Convolutional layers:
   Apply filters to the image to detect features such as edges, lines, textures and abnormal regions.

   Filters/Kernels:
   Small matrices that slide over the image and detect specific patterns. Different filters learn different features.

   Activation function — ReLU:
   Introduces non-linearity and helps the network learn complex patterns.

   Pooling layers:
   Reduce the spatial size of feature maps while retaining important information. Max pooling is commonly used.

   Fully Connected layers:
   Combine the extracted features and perform the final classification.

   Output layer:
   Produces the probability of the image belonging to each class.

3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.
   Answer:
   In a fully connected network, every input pixel is connected to every neuron. This creates a very large number of parameters.
   In CNNs, the **same filter is reused across different locations** of the image. This is called **parameter sharing**.
   For example, a 3×3 filter contains only 9 weights plus a bias, and the same weights are applied across the complete image.
   Therefore:

- Number of trainable parameters is greatly reduced.
- Training becomes faster.
- Memory requirements are reduced.
- The network can detect the same feature at different image locations.

Thus, parameter sharing makes CNNs much more efficient than fully connected networks for image data.

4. Discuss suitable **regularization techniques such as dropout, data augmentation and batch normalization** to reduce overfitting.
   Answer:
   Medical image datasets can cause overfitting because the model may memorize training images instead of learning general patterns.
   Suitable techniques include:
   1.Data Augmentation:
   Generate variations of training images using operations such as small rotations, translations, cropping and brightness changes. This improves generalization.
   2. Dropout:
   Randomly disables some neurons during training. This prevents the network from depending too heavily on particular neurons.
   3. Batch Normalization:
   Normalizes activations during training, making learning more stable and often improving convergence.
   4. Early Stopping:
   Stops training when validation performance stops improving.
   These techniques help the model perform better on unseen X-ray images.

5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.
   Answer:

Feature	**AlexNet**	**ResNet**
Depth	Relatively shallow	Much deeper
Feature learning	Good	Excellent

Feature	AlexNet	ResNet
Training	Easier	More complex but supported by residual connections
Vanishing gradient	More likely in deeper extensions	Reduced using skip connections
Parameters	Relatively high for its depth	Deeper models can be parameter-efficient
Performance	Good baseline	Usually better for complex image classification
Medical image application	Suitable for simple systems	More suitable for high-accuracy systems

Recommendation: ResNet
For medical X-ray classification, **ResNet** is more appropriate because it can learn complex hierarchical features from X-ray images. Its **residual/skip connections** help solve the vanishing-gradient problem and allow deeper networks to be trained effectively.
A pretrained ResNet can also be fine-tuned on the medical X-ray dataset, reducing training requirements.
Therefore, **ResNet is preferred when classification accuracy and strong feature learning are important**, although a smaller ResNet model can be selected if computational resources are limited.

Case Study Question 2: CNN-Based Autonomous Vehicle Object Recognition (10 Marks)
An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.
Question:

1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.
   Answer:
   A CNN can be designed to process camera images and identify objects such as:

- Cars
- Pedestrians
- Bicycles
- Traffic signs
- Road obstacles

A typical architecture is:
Input Image → Convolution → ReLU → Pooling → Convolution → ReLU → Pooling → Deeper Convolution Layers → Fully Connected/Detection Head → Output

The input camera image is resized to a fixed resolution. The CNN progressively extracts features and finally predicts the object category.
For a real autonomous vehicle, an object-detection architecture based on CNNs would generally be preferred because the system must identify **both the object and its location**.

2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional **filters progressively learn low-level and high-level features**.
   Answer:
   Input Layer:
   Receives camera images and converts them into a suitable size and format.
   Convolutional Layers:
   Apply filters to detect visual features.
   Activation Function:
   ReLU introduces non-linearity and allows the network to learn complex patterns.
   Pooling Layers:
   Reduce feature-map dimensions and computational requirements while retaining important features.
   Fully Connected/Output Layers:
   Use learned features to classify objects or produce detection predictions.
   Progressive Feature Learning
   CNN filters learn features progressively:
   Early layers:
   Edges, corners and simple shapes.
   ↓
   Middle layers:
   Textures, curves, wheels, signs and object parts.
   ↓
   Deep layers:
   Complete objects such as cars, pedestrians and bicycles.
   Thus, deeper layers contain increasingly meaningful representations of the road scene.

3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.
   Answer:
   CNN filters use the **same weights at different locations** in an image.
   For example, a filter that learns a vertical edge can detect that edge whether it appears on the left, center or right side of the image.
   This provides:

- Fewer trainable parameters.
- Lower memory requirements.
- Reduced computational complexity.
- Faster training.
- Better detection of features at different image positions.

Parameter sharing is particularly useful for autonomous vehicles because objects can appear at different locations in camera frames.

4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.
   Answer:
   The model may overfit because road images can have repeated backgrounds, vehicles and environmental patterns.
   Suitable techniques are:
   Data Augmentation:
   Use changes in brightness, contrast, cropping, rotation and image position to create diverse training examples.
   Dropout:
   Randomly disables neurons during training to reduce dependency on particular features.
   Batch Normalization:
   Normalizes intermediate activations and helps stabilize training.
   Weight Regularization:
   L1 or L2 regularization can penalize excessively large weights.
   Early Stopping:
   Stops training when validation performance begins to decrease.
   These methods improve generalization to new roads and weather conditions.

5. Compare **AlexNet and ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.
   Answer:

Feature	AlexNet	ResNet
Architecture	Traditional CNN	CNN with residual/skip connections
Depth	8 major layers	Can be much deeper
Training difficulty	Relatively simple	Easier to train deep networks due to skip connections
Vanishing gradients	More concern as networks become deeper	Significantly reduced
Feature learning	Good	Stronger hierarchical feature learning
Parameters	Large for its era	Better efficiency in deeper architectures
Accuracy	Good baseline	Generally higher

6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.
   Answer:
   ResNet is the better choice for autonomous vehicle object recognition.
   Reasons:

1. **Higher accuracy:**
   It can learn complex features from road scenes.
2. **Deep feature learning:**
   It learns features ranging from edges to complete objects.
3. **Residual connections:**
   Skip connections reduce vanishing-gradient problems.
4. **Better generalization:**
   It can handle variations in lighting, distance and viewing angle.
5. **Scalability:**
   Different ResNet sizes can be selected according to available hardware.
   However, autonomous vehicles require **real-time processing**, so a very large ResNet may be computationally expensive. A smaller ResNet variant or optimized version should be selected when low latency is required.

   Final Recommendation
   ResNet is preferred over AlexNet because it provides better feature learning and classification performance while solving the training difficulties associated with deeper networks. For an actual vehicle, the selected ResNet model should be optimized to balance **accuracy, computational cost, memory usage and real-time inference speed**.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand